How Does Schema Complexity Predict Performance in Training-Free LLM Information Extraction?

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Abstract

Training-free LLM information extraction degrades as schema complexity grows, yet prior work conflates two factors: type count and type confusability. We present what is, to our knowledge, the first controlled study isolating these factors across NER, RE, and event detection on four datasets. We quantify confusability via the Schema Confusability Score (SCS) and its batch-level aggregation, the Batch Confusability Score (BCS). Degradation operates along two dimensions with task-dependent effect structures: count-related recall decline (up to -49 pp on DocRED 96-relation RE) and confusability-related type substitution. In RE, BCS explains 22–58% of cross-batch accuracy variance (FewRel/DocRED), independent of output format (paired constrained/unconstrained comparison yields identical correlations); NER shows no batch-level BCS signal, as errors scatter across missed entities and boundary failures. Schema decomposition fails due to merge amplification (NER, ED) and routing failure (RE). Both dimensions replicate across four models spanning three architecture families (Qwen, LLaMA, Gemini; 8B–70B), two decoding formats, and four prompt designs.

1 Introduction

Training-free[1][2] LLM information extraction (prompting a large language model with a target schema and asking it to extract structured information without any task-specific fine-tuning) enables rapid deployment across new domains and ontologies. This zero-shot paradigm eliminates the need for labeled data, and is attractive for applications where annotation is expensive or schemas evolve frequently (Xu et al., 2024). Yet the paradigm carries a critical limitation: performance degrades sharply as the number of candidate types in the prompt schema grows. On named entity recognition, recall drops from 0.72 with a single candidate type to a plateau of $\sim$0.27 across intermediate schema sizes (22–44 types) before partial recovery at full-schema. On relation extraction, accuracy falls from 85.7% with one relation to 49.5% with 80 relations (sentence-level) and from 88.0% to 38.7% with 96 relations (document-level), near-halving or worse. This degradation is not academic: real-world ontologies routinely define 50–200+ types, which places them squarely in the regime where training-free extraction becomes unreliable.

What explains this degradation? Prior decomposition strategies (Zhou et al., 2024; Liu et al., 2025; Shrimal et al., 2025) conflate two factors: having more types to discriminate among (count) and having semantically similar types to confuse (confusability) (hereafter Dimension 1 and Dimension 2). None isolates which factor predicts the gains (§2).

This study combines correlational analyses with quasi-experimental manipulation (BCS-controlled batches) to characterize how schema properties predict extraction accuracy.

We ask three questions. **RQ1**: Does type confusability independently predict extraction degradation beyond the effect of type count? **RQ2**: Through what modes do count and confusability each degrade performance? **RQ3**: Can confusability-aware schema decomposition improve extraction quality?

To our knowledge, we present the first controlled study isolating type confusability from type count via stratified design in zero-shot LLM IE, spanning NER, RE, and event detection (ED). We quantify confusability using pairwise embedding similarity, the *Schema Confusability Score* (SCS), and its batch-level aggregation, the *Batch Confusability Score* (BCS). We evaluate on Few-NERD

[1] We use "training-free" and "zero-shot" interchangeably to denote prompting without parameter updates.

[2] Code and data will be released upon acceptance.

(Ding et al., 2021), FewRel (Han et al., 2018), DocRED (Yao et al., 2019), and MAVEN (Wang et al., 2020b) using Qwen3-32B (§3).

Our experiments yield four findings: (1) Degradation operates along two dimensions with bounded interaction: count-related recall decline (up to -49 pp) and confusability-related type substitution, confirmed on sentence- and document-level RE (§4.1, §4.4). (2) Pairwise SCS predicts type confusion across NER, RE, and ED (§4.2); at batch level, BCS predicts RE accuracy independent of output format (§4.3) but not NER accuracy; per-type aggregation is confounded by centrality on homogeneous schemas (§4.5). (3) Effects are consistent across schema sizes, prompt designs, and models from three architecture families (§4.3). (4) Schema decomposition fails due to merge amplification (NER, ED) and routing failure (RE), despite oracle routing recovering substantial accuracy (§5). Several findings are non-obvious: modest pairwise signals ($r^2 \approx 1.3\%$) amplify to $R^2 = 22$–58% at batch level; renaming confusable types *decreases* accuracy (Appendix X); and few-shot demonstrations widen rather than close the confusability gap.

Our contributions are:

1. We present what is, to our knowledge, the first controlled study that isolates count and confusability effects in training-free LLM extraction. Building on the embedding-similarity criterion from prior schema-partitioning work (Liu et al., 2025), we design stratified experiments that quantify confusability's predictive relationship with extraction degradation across NER, RE, and event detection (§4.1, §4.2, §4.3).

2. We document a task asymmetry: BCS explains 22–58% of RE accuracy variance through type substitution—an effect confirmed as format-independent—yet shows no batch-level signal in NER despite significant pairwise effects (§4.3, §4.4).

3. We identify structural barriers to confusability-aware extraction: schema decomposition fails due to merge amplification (NER, ED) and routing failure (RE), establishing that the decompose-then-merge paradigm faces fundamental precision–recall tradeoffs in training-free IE (§5).

2 Related Work

Training-free LLM IE. Training-free LLM IE spans instruction tuning (Wang et al., 2023), code-style schema encoding (Li et al., 2024, 2023b), annotation-guided extraction (Sainz et al., 2024; Qi et al., 2024), schema-driven decoding (Shen et al., 2025; Luo et al., 2025), and constrained JSON decoding (Geng et al., 2025; Willard and Louf, 2023). For NER, methods include zero-shot decomposition (Xie et al., 2023; Ashok and Lipton, 2023), distillation (Zhou et al., 2024), bidirectional encoders (Zaratiana et al., 2024, 2025), and fine-grained entity typing (Gillick et al., 2014). For RE, recent work examines zero-shot relation triplet extraction via prompting (Chia et al., 2022), constrained generative extraction (Josifoski et al., 2022), revisits relation extraction in the LLM era (Wadhwa et al., 2023), and zero-shot RE via shared embedding spaces (Boylan et al., 2025). These methods perform well with moderate schemas (4–20 types) but none investigates how performance degrades as schema complexity scales, nor proposes quantitative metrics for inter-type confusability. Wei et al. (2023) reformulate zero-shot IE as multi-turn QA with ChatGPT but do not analyze schema-size effects.

Schema complexity in IE. The most directly related work is LLM-PEE (Liu et al., 2025), which partitions large event schemas by type embedding similarity. Our work differs in three ways: (1) we conduct controlled analysis to understand *why* extraction degrades rather than developing an engineering solution; (2) our BCS-controlled experiments isolate count from confusability, which LLM-PEE conflates; (3) our SDGD experiments reveal that merge amplification (not partitioning quality) is the bottleneck in decompose-then-merge pipelines. LLM-PEE's union merge suffers the same precision collapse (0.56→0.21; §5). In short, LLM-PEE assumes decomposition helps; we show why it should (confusability), why it does not (merge amplification), and what predicts the degradation (BCS). Other approaches address schema complexity via hard negative mining (Gui et al., 2024), schema definition optimization (Shrimal et al., 2025), multi-agent debate (Lu et al., 2025; Wang et al., 2025), schema-aware decomposition for event extraction (Shiri et al., 2024), and active prompting (Zhao et al., 2026); all treat it as an engineering problem rather than isolating the underlying degradation factors. CrossAgentIE's cross-type

debate mechanism addresses confusability through multi-agent deliberation rather than batch composition, a complementary approach that warrants empirical comparison. Concurrently, Chen et al. (2026) show that instance count degrades LLM multi-instance processing more than context length, consistent with our count-dominant finding. Concurrently, Vandemoortele et al. (2025) reduces label spaces via iterative LLM filtering, Li et al. (2025) caches schemas for on-device extraction, and Liang et al. (2025) treats schemas as parameterized tools. Our framework explains *why* such reduction works: it implicitly lowers batch confusability.

Schema complexity in dialogue and IE. The Schema-Guided Dialogue dataset (Rastogi et al., 2020) and SGD-X (Lee et al., 2022) document performance degradation with schema linguistic variation. Li et al. (2023a) observe count-dependent degradation in ChatGPT IE. Prompt order sensitivity (Lu et al., 2022) and label ordering effects are well documented; our prompt-variant experiments (§H.1) control for such artifacts. Volkanovska (2025) categorize LLM NER errors including type confusion. The relationship between label similarity and confusion has precedent: pairwise confusion regularization improves fine-grained visual recognition (Dubey et al., 2018), correlated label noise captures inter-class confusability (Collier et al., 2021), hierarchical classification suffers sibling confusion (Deng et al., 2010), and label confusion matrices improve text classification (Guo et al., 2021; Zhou et al., 2025; Müller et al., 2019; Peterson et al., 2019). Hierarchical label sets (Novack et al., 2023), enriched type descriptions (Cocchieri et al., 2025), and universal entity taxonomies (Yang et al., 2025) address related problems but are orthogonal to confusability-aware batching.

SCS novelty. SCS shares its formal definition (pairwise cosine similarity of type name embeddings) with the partitioning criterion in LLM-PEE (Liu et al., 2025). While LLM-PEE uses this criterion for engineering optimization (minimizing cross-batch confusion during decomposition), our work uses the same metric for scientific analysis (isolating confusability's contribution to degradation under controlled conditions). The use of embedding distance for type discrimination has parallels in metric learning (Snell et al., 2017) and zero-shot classification (Yin et al., 2019). Our contribution is the controlled experimental framework that isolates confusability's predictive relationship with extraction degradation; the modest pairwise r^2 (1.3–1.7%) compounds at the batch level to explain 22–58% of cross-batch RE variance (§4.3). Our centrality confound (§4.5) connects to representation degeneration (Gao et al., 2019; Ethayarajh, 2019): local clusters can be isotropic (Cai et al., 2021), explaining why MAVEN's single-ontology types conflate confusability with centrality.

3 Experimental Setup

We evaluate how prompt schema size and inter-type confusability affect zero-shot LLM extraction across three information extraction tasks and four datasets: named entity recognition (NER), sentence-level relation extraction (RE), document-level RE, and event detection (ED).

3.1 Tasks and Datasets

Few-NERD (NER). Few-NERD (Ding et al., 2021) provides 66 fine-grained entity types organized into 8 coarse categories (person, location, organization, building, art, product, event, other). The dataset spans diverse domains and is suitable for cross-domain evaluation. We sample 1,000 test instances uniformly across test instances; type frequency follows the natural long-tail distribution of Few-NERD. Evaluation uses token-level text matching. Pairwise analysis covers 1,953 type pairs with non-zero co-occurrence out of $\binom{66}{2} = 2{,}145$ possible pairs.

FewRel (RE). FewRel (Han et al., 2018) contains 80 relation types sourced from Wikidata. Since the validation split covers only 16 relations, we combine the training and validation splits to obtain the full 80-relation pool required for schema-scale experiments, and sample 1,000 balanced instances (12–13 per relation type, stratified to ensure equal representation). A held-out validation confirms SCS–confusion generalization across the split boundary (Appendix M). Each instance provides a sentence with marked head and tail entities; the task is to classify the relation. FewRel's balanced design (700 instances per relation type) eliminates type frequency as a potential confound; this balance makes it the primary dataset for BCS analysis.

DocRED (Document-level RE). DocRED (Yao et al., 2019) contains 96 relation types sourced from Wikidata, evaluated at the document level where entity pairs may span multiple sentences. We sample 1,000 instances balanced across 96 relation types

Dataset	Task	# Types	# Eval Units	Domain	BCS Range	Embedding
Few-NERD	NER	66	1,000	Cross	0.77–0.93	Qwen3-32B
FewRel	RE (sent.)	80	1,000	Cross	0.42–0.97	MiniLM
DocRED	RE (doc.)	96	1,000	Cross	0.53–0.73	MiniLM
MAVEN	ED	168	18,904†	Same	0.87–0.97	MiniLM

Table 1: Dataset statistics. BCS Range: min–max over count-curve batches. Embedding: model used for SCS/BCS; ranges are not comparable across embedding models. †Trigger mentions across 8,042 sentences.

(∼10 per type) from the DocRED development set and formulate the task identically to FewRel: given a document with marked head and tail entities, classify the relation. DocRED extends the RE evaluation to a larger schema (96 vs. 80 types) and document-level context, testing whether confusability effects persist under longer input contexts and cross-sentence reasoning.

MAVEN (ED). MAVEN (Wang et al., 2020b) defines 168 event types derived from the FrameNet ontology; all types belong to a single ontology (FrameNet), so this is a same-domain schema. We use the validation set (8,042 sentences containing 18,904 event mentions) for evaluation, as test set labels are not publicly available. The task requires both trigger identification and event type classification. MAVEN serves a supplementary role because its same-domain schema produces centrality confounds at the batch level (§4.5).

Table 1 summarizes the dataset statistics, including the Schema Confusability Score (SCS) ranges defined in §3.2.

3.2 Schema Confusability Score (SCS)

We define the Schema Confusability Score (SCS) to quantify semantic overlap between types in a schema. Given a type t with name string $\text{name}(t)$, an embedding function $\text{embed}(\cdot)$ maps the type name to a vector representation. The **pairwise SCS** between types t_i and t_j is:

$$\text{SCS}(t_i, t_j) = \cos\big(\text{embed}(\text{name}(t_i)),\ \text{embed}(\text{name}(t_j))\big)$$

For a batch of types $B = \{t_1, \ldots, t_k\}$, the **Batch Confusability Score (BCS)** aggregates all pairwise similarities:

$$\text{BCS}(B) = \frac{1}{\binom{k}{2}} \sum_{i<j,\ t_i,t_j \in B} \text{SCS}(t_i, t_j)$$

We use two embeddings to provide complementary perspectives: Qwen3 captures the extraction model's own type representations (self-similarity), while MiniLM provides an independent general-purpose baseline. Cross-embedding consistency holds on all datasets (Appendix I); MiniLM consistently yields equal or stronger correlations, which confirms that Qwen3-based estimates are conservative. We compute SCS with: (1) Qwen3-32B (Qwen Team, 2025) hidden-state representations (5,120-d) and (2) all-MiniLM-L6-v2 (Wang et al., 2020a; Reimers and Gurevych, 2019), a 384-d sentence encoder. Few-NERD uses Qwen3 because its fine-grained same-category pairs (e.g., "person-artist" vs. "person-actor") require the higher-dimensional space. MiniLM is the primary BCS embedding because sentence-transformers are trained for semantic similarity (Reimers and Gurevych, 2019); Qwen3 causal-LM states show compressed BCS range (std $= 0.034$ vs. MiniLM std $= 0.081$), consistent with autoregressive anisotropy (Ethayarajh, 2019). Name-only SCS yields equivalent predictive accuracy to name+definition (Δr: -0.023 to $+0.035$; Appendix M).

SCS as a lightweight proxy. SCS is a pre-hoc proxy computable without running extraction. Pairwise SCS correlates with observed type confusion (Few-NERD $r = 0.112$, FewRel $r = 0.187$, both $p < 0.001$; Table 2), though context, frequency, and definition quality also contribute (§4.2).

3.3 Experimental Design

We conduct two complementary experimental paradigms.

Experiment 1: Count Curves. We vary the number of types presented in the prompt schema to measure performance as a function of schema size. For each count level, we sample 25 random type batches for Few-NERD and FewRel (5 for DocRED and MAVEN)[3] and evaluate the same 1,000 test samples across all batches. At count $= 1$, every type is evaluated individually as a solo baseline. The remaining count levels follow natural partition points in each schema's type hierarchy rather than uniform spacing: {2, 6, 12, 22, 44, 66} for Few-NERD, {4, 8, 16, 32, 40, 64, 80} for FewRel, {4, 8, 16, 32, 64, 96} for DocRED, and {6, 12, 22, 44, 64} for MAVEN.

Experiment 2: BCS-Controlled. We fix type count and vary BCS to isolate confusability from schema size. For each count level, we construct one

[3]DocRED and MAVEN use fewer batches due to higher per-batch evaluation cost (document-level RE and 8,042 sentences, respectively).

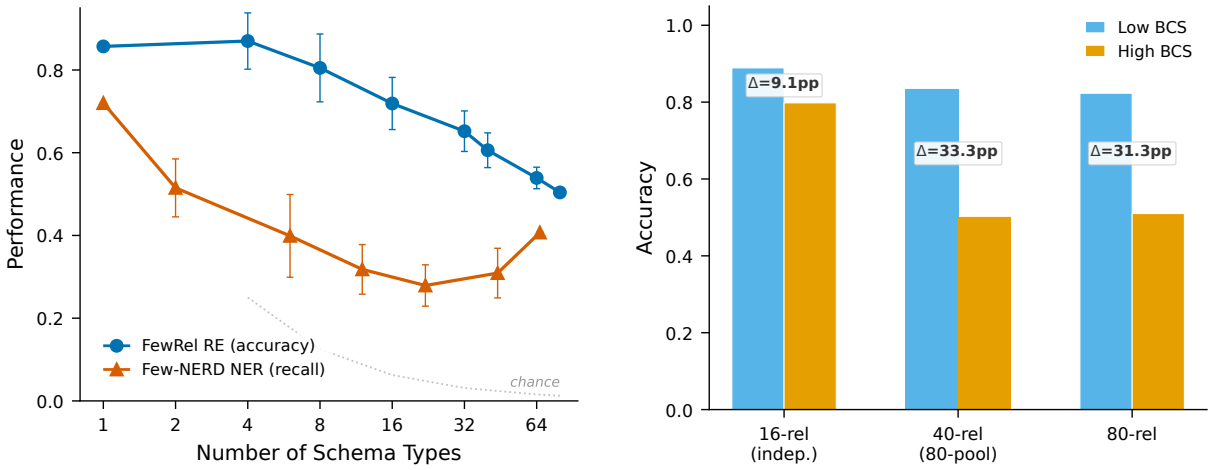


Figure 1: **(a)** Count-degradation curves. Error bars: ± 1 SD across 3 seeds $\times$ 25 batches. RE declines monotonically; NER shows a U-shape. Rightmost points are full-schema (negligible sampling variance). **(b)** BCS-controlled FewRel accuracy. Slopes are consistent across the independent and within-pool 16-relation conditions ($p = 0.60$); wider BCS range at 80 relations produces larger gaps.

low-BCS batch, one high-BCS batch, and 25 random batches (27 total). Extreme batches are constructed via greedy search over the pairwise SCS matrix: the low-BCS batch starts from the least-similar type pair and iteratively adds the type with lowest maximum similarity to already-selected types; the high-BCS batch mirrors this with maximum similarity. We evaluate on Few-NERD (count $\in \{6, 12\}$), FewRel (count $\in \{8, 16\}$, both 16-relation (pilot) and 80-relation pools), and DocRED (count $\in \{16, 32\}$). Within-pool 16-relation subsets from the 80-relation pool isolate confusability from pool composition. MAVEN results are reported separately due to a centrality confound (§4.5).

3.4 Evaluation Metrics

NER: text-match F1, where a prediction is correct if the exact text string matches after lowercasing and whitespace normalization, *and* the type matches; partial spans count as incorrect. **RE**: accuracy (exact match of predicted relation label to gold). Each batch evaluates only samples whose gold relation is included in the candidate set; out-of-schema samples are excluded before inference. The prompt allows a `none` response for cases where the model judges no candidate applies; any such prediction is scored as incorrect since the gold is always among the candidates by construction. **ED**: relaxed F1, where any token overlap between predicted and gold trigger spans, with correct event type.

3.5 Model and Inference

We use Qwen3-32B-FP8 (Qwen Team, 2025) served via vLLM v0.20.2 (Kwon et al., 2023) on a single NVIDIA RTX PRO 6000 GPU (96 GB) with greedy decoding (temperature $= 0$, seed 42). Two NER/ED prompt versions are evaluated: v1 requests character offsets (evaluated via text matching); v2 adds format constraints that double precision with minimal recall loss. RE uses a single prompt version throughout. Both versions produce consistent degradation patterns; we report both where applicable to quantify formatting sensitivity (Sclar et al., 2024) (templates in Appendix G). Cross-model validation uses Meta-Llama-3.1-70B-Instruct (FP8) and Meta-Llama-3.1-8B-Instruct (Dubey et al., 2024). All experiments use single-seed evaluation with greedy decoding, which substantially reduces model-level randomness (residual SD = 0.88 pp at count 80 under greedy decoding, likely attributable to FP8 quantization non-determinism); bootstrap CIs capture the remaining sampling variance from batch composition. Pair-level CIs use percentile bootstrap (B=10,000); batch-level CIs use B=1,000.

4 Results

Unless otherwise noted, all experiments in this section use the v1 prompt (§3.5; full templates in Appendix G).

Dataset	Embed.	n pairs	r	r^2	95% CI	p
Few-NERD	Qwen3	1,953	0.112	0.013	[0.076, 0.156]	<0.001
MAVEN	MiniLM	11,026	0.129	0.017	[0.104, 0.156]	<0.001

Table 2: Pairwise SCS–confusion correlation. Confusion rate is bidirectional $C(t_i \to t_j) + C(t_j \to t_i)$ divided by gold frequency sum; CIs are percentile bootstrap (B=10,000). FewRel ($r = 0.187$; §3.2) excluded (per-instance RE format). Type-level cluster bootstrap CIs exclude zero (Few-NERD [0.069, 0.219]; MAVEN [0.081, 0.185]).

4.1 Count-Degradation Curves

Figure 1a shows RE accuracy declines monotonically with schema size (0.86→0.50 for FewRel). NER recall declines through intermediate counts (0.51→0.27, plateauing across counts 22–44) before partial recovery at full-schema (0.41), where the elimination of type-filtering overhead restores extraction performance. DocRED replicates the monotonic RE pattern (88.0%→38.7%, −56%; Appendix). MAVEN exhibits a reversed trend attributable to centrality confounding (§4.5). The effect replicates on Llama-3.1-8B-Instruct (−43 pp on Few-NERD; per-batch $r = 0.725$; Table 3). The pattern is stable across three seeds (max $\Delta =$ 10.7 pp, within 1 SD; intra-batch variance exceeds inter-seed variance by >5×; Appendix R). Formal trend analysis confirms monotonic RE decline (Jonckheere-Terpstra $J = 109{,}707$, $p < 0.001$). Count curves establish trend direction; magnitude estimation relies on the higher-powered BCS-controlled experiments (§4.3).

4.2 Pairwise Confusability and Type Confusion

We test whether pairwise SCS predicts type-level confusion rates. For each type pair (t_i, t_j), we compute the confusion rate (the fraction of t_i gold instances misclassified as t_j) and correlate it with $\text{SCS}(t_i, t_j)$. Table 2 reports Pearson correlations with bootstrap 95% CIs.

The correlations are statistically significant but modest: $r^2 \approx 1.3$–1.7%. Type-level cluster bootstrap yields wider but still significant CIs (Few-NERD: [0.069, 0.219]; MAVEN: [0.081, 0.185]; $B = 10{,}000$). Spearman correlations corroborate: $\rho = 0.144$–0.227 across all pairs, strengthening to $\rho = 0.27$–0.31 among pairs with nonzero confusion (though conditioning on the dependent variable may inflate this).

The remaining variance reflects entity frequency, prompt position (Liu et al., 2024), mention ambiguity, and definition quality. While pairwise r^2 is modest, correlated type pairs within batches compound this signal: batch-level BCS explains 22–58% of accuracy variance (§4.3). The independent MiniLM embedding yields a *stronger* SCS–confusion correlation on Few-NERD ($r = 0.172$ vs. Qwen3 $r = 0.138$; Appendix), which indicates that shared-representation bias is conservative. SCS outperforms simpler baselines (Levenshtein, WordNet Wu-Palmer, same-parent; Appendix P). Appendix W lists the most confusable type pairs with concrete examples.

4.3 BCS-Controlled Experiments

BCS explains 22–58% of cross-batch RE accuracy variance; confusability is a substantial predictor alongside count. We fix type count and vary BCS to isolate this effect. FewRel's balanced design (700 instances per type) eliminates type frequency as a confound; the within-pool analysis (§4.3) controls for pool composition.

In BCS-controlled FewRel subsets, the high-BCS quartile shows $\Delta = -5.3$ pp lower accuracy than low-BCS (quartile comparison on 100 random batches; based on a single extreme batch per BCS level at 16-rel scale; full data in Appendix J). At 80 relations, extreme-batch construction amplifies this to −31.2 pp (95% CI [17.7, 42.7]), stable across count levels. An intermediate count $= 40$ evaluation (40-relation subsets sampled from the 80-relation pool) yields consistent gaps (−33.3 pp at count 8, −31.8 pp at count 16) and confirms monotonic decline.

Document-level RE (DocRED). The effect replicates on DocRED's 96-relation schema (n=55 per count level): extreme-BCS gap −25.0 pp at count 16 ($\rho = -0.762$) and −15.7 pp at count 32 ($\rho = -0.693$; both $p < 0.001$). Pooled across counts, BCS explains 58% of cross-batch variance ($R^2 = 0.578$, $n = 110$; 95% CI [0.46, 0.68]; per-count: $R^2 = 0.653$ at count 16, 0.614 at count 32), substantially higher than FewRel ($R^2 = 0.218$).

Within-pool controlled comparison. Within-pool analysis confirms slope consistency across pools (permutation $p \approx 0.60$; Appendix L). The 80-relation extreme-batch gap (§4.3) reflects wider BCS range, not stronger per-unit sensitivity.

Practical magnitude. Direct regression on 100 random 16-type batches from the 80-relation pool shows BCS explains 22% of cross-batch accuracy

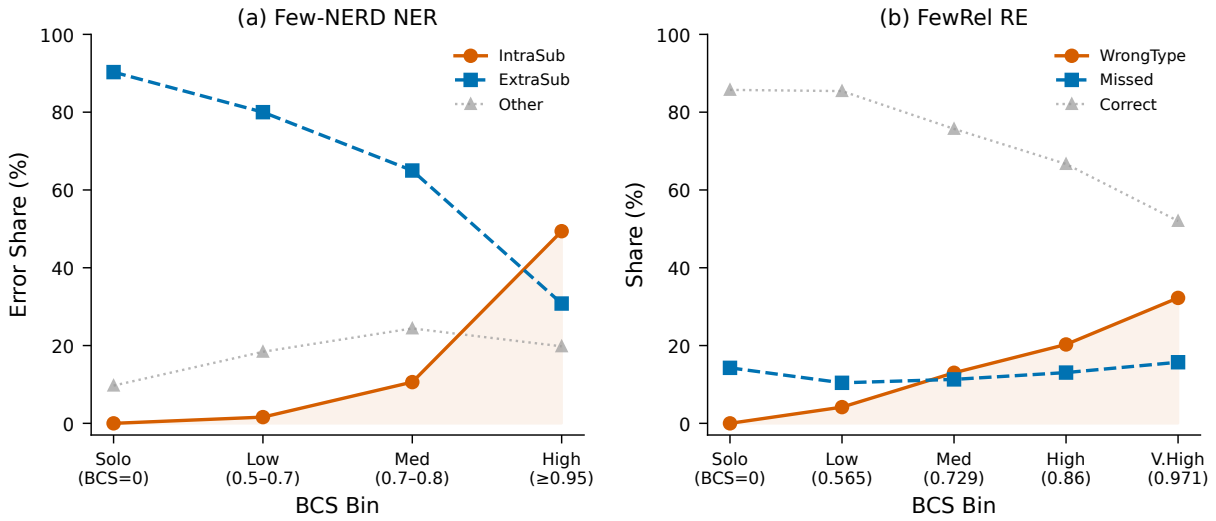


Figure 2: Error taxonomy decomposition (Few-NERD NER and FewRel RE). IntraSub scales monotonically with BCS across all evaluated datasets ($\rho = 0.84$–1.0; Table 5 includes DocRED). RE missed rates remain flat (10–16%), but NER missed rates are substantially higher (43–65%) and non-monotonic.

variance ($R^2 = 0.218$, $p < 0.001$), scaling to 58% on DocRED (pooled, $n = 110$). The typical BCS span (IQR = 0.054) yields an expected accuracy gap of 2.2 pp (95% CI [0.4, 4.0]). Quartile comparison confirms: bottom-quartile BCS batches average 82.8% vs. 78.9% for top-quartile ($\Delta = 3.9$ pp). A post-hoc Monte Carlo simulation confirms the $\sim 3\times$ pair-to-batch amplification (Appendix N).

Cross-model validation. The effect replicates on Meta-Llama-3.1-8B-Instruct: extreme-BCS gaps of -27.1 pp (count 8) and -23.9 pp (count 16), reaching 77–87% of Qwen3-32B's gap. Both models converge to ≈51% under high confusability, suggesting a task-inherent difficulty floor (Table 4). Gemini 2.5 Flash replicates Dimension 2 at batch level: low-BCS 90.5% vs. high-BCS 72.9% ($\Delta = -17.6$ pp, 95% CI [15.1, 20.1], $n = 50$) at count 8, with the smaller gap likely reflecting ceiling compression from Gemini's higher baseline. Llama-3.1-70B-Instruct (FP8) confirms directional consistency (Appendix T, U). LLaMA-3.1-8B replicates the BCS effect on FewRel RE ($\Delta = -27.1$ pp) but shows no batch-level signal on NER, consistent with the task asymmetry observed across all model sizes.

Prompt sensitivity. Under the v2 prompt, the extreme-batch BCS gap is -25.5 pp ($r = -0.919$, $p < 0.001$; n=25), modestly attenuated from v1 but highly significant. NER shows no batch-level BCS–F1 correlation under either prompt ($r \approx 0$, $n = 450$; §4.4). The effect holds across four structurally distinct prompt variants at count 32 (all $r \leq -0.47$; Appendix H.1), ruling out prompt design as the source. Few-shot demonstrations widen the confusability gap (+4.3 pp under 5-shot), though this finding is limited to a single dataset and count level (Appendix V).

4.4 Multi-Dimensional Degradation

Error taxonomy decomposition (Table 5 and Figure 2) shows intra-batch substitution (IntraSub) scales monotonically with BCS across all three datasets: $\rho = 1.0$ ($p < 0.001$) for Few-NERD NER, $\rho = 0.954$ ($p < 0.001$) for FewRel sentence-level RE, and $\rho = 0.843$ ($p < 10^{-30}$, $n = 110$) for DocRED document-level RE. The slightly lower DocRED correlation reflects additional noise from document-level context and cross-sentence reasoning. In NER, missed rates are higher (43–65%) and non-monotonic; confusable types impair entity detection itself. In DocRED, IntraSub accounts for 72% of errors in high-BCS batches vs. 15% in low-BCS batches; ExtraSub is zero (forced-choice format). Extreme-BCS batches produce 9.9–27.5$\times$ more confusion than low-BCS batches.

These results establish two dimensions: **Dimension 1** (count predicts recall decline) and **Dimension 2** (confusability predicts type confusion). The full-range regression ($n = 69$, five count levels) serves as the primary separability test; the 2×2 factorial ($n = 54$) as a sensitivity check. We select the equivalence margin via convergent evidence: (i) an interaction exceeding the primary BCS ef-

fect (IQR-predicted $\Delta = 2.2$ pp; §4.3) would represent $\geq 100\%$ amplification, fundamentally altering whether the dimensions are independently addressable; (ii) Cohen's $d = 0.5$ (medium effect) at the within-cell SD of 4.6 pp yields ± 2.3 pp. Both converge on ± 2–3 pp. TOST (Lakens, 2017; Schuirmann, 1987) on FewRel bounds the interaction at ± 3 pp ($p = 0.003$); at the stricter ± 2 pp margin, equivalence is marginal ($p = 0.047$) and inconclusive on the logit scale ($p = 0.209$; Appendix C). On DocRED (n=55 per count level), the raw factorial interaction is -9.4 pp (95% CI $[-12.9, -6.0]$), but this is confounded by narrower BCS ranges at count 32 (spread 0.26 vs. 0.38). Restricting both count levels to the overlapping BCS range [0.43, 0.70] reduces the interaction from 7.52 pp (CI [1.49, 12.83]) to 0.93 pp (CI $[-7.13, 7.44]$)—consistent with bounded interaction, though the reduced sample ($n = 29{+}55$) has 80% power only for effects ≥ 10.5 pp (Appendix C). In NER, batch-level BCS shows no predictive signal (§4.3), where NER errors distribute across miss, boundary, and type modes rather than concentrating in type substitution (relaxed matching confirms this null is not a metric artifact; Appendix Y); the factorial test is inconclusive ($p = 0.50$).

4.5 Centrality Confound on Homogeneous Schemas

Per-type analysis on MAVEN reveals an anomalous *positive* BCS–F1 correlation ($\rho = +0.184$–0.208, $p < 0.03$): FrameNet-derived types cluster tightly (Cai et al., 2021), conflating confusability with centrality. Controlling for log frequency and degree centrality, the correlation vanishes ($p > 0.2$; Appendix E), though the pairwise SCS signal remains intact ($r = 0.129$, $p < 0.001$). Few-NERD's cross-domain schema shows the expected negative direction ($\rho = -0.293$, $p = 0.02$); mean BCS is reliable only for cross-domain schemas.

5 Schema Decomposition: Failure, Diagnosis, and Mitigation

A natural intervention is to partition the schema so confusable types never co-occur. We test **Schema-DAG-Guided Decomposition** (SDGD), which applies greedy graph coloring on the SCS graph to assign types to low-confusability batches.

NER and ED: merge amplification. All decomposition modes fall below the single-pass NER baseline (F1 = 0.445): SDGD obtains 0.251, random batching 0.320. The core failure is *merge amplification*: presenting text k times collapses precision (0.563→0.210–0.249). MAVEN event detection replicates this (precision 0.287→0.091; Appendix B).

RE: routing failure. Oracle routing recovers accuracy (0.802 vs. 0.504 single-pass; Table 10), but confusability-aware grouping yields only +1.4 pp over random ($p \approx 0.19$, n.s.). All practical routing strategies fall below the single-pass baseline (Appendix B.1).

6 Discussion and Conclusion

Boundary conditions. The two-dimensional decomposition applies to cross-domain schemas; same-domain schemas exhibit centrality confounding (§4.5). Approximate separability is confirmed on FewRel and DocRED (§4.4); logit-scale evidence remains inconclusive. Both dimensions replicate across four models (8B–70B), two decoding formats, four prompt designs, and two RE datasets.

Embedding similarity and format independence. SCS captures semantic proximity in LLMs' type-resolution space, amplifying from pairwise $R^2 = 1.3\%$ to batch-level 18–25% via correlated aggregation (Appendix N). The BCS–performance relationship holds across all RE settings but not NER; a paired format comparison confirms format independence (Appendix K). The task asymmetry reflects error-mode structure: RE errors concentrate in type substitutions (Tam et al., 2024), whereas NER errors scatter across miss, boundary, and type modes (Schall and de Melo, 2025). SCS likely proxies deeper conceptual overlap persisting despite label renaming (Appendix X); we hypothesize high-SCS types intensify softmax competition—testable via logit-space analysis.

Practical implications. For NER, reducing candidate types per prompt is the primary lever; for RE, we suggest flagging type pairs with SCS > 0.85 and monitoring batch-level BCS (typical gap 2–4 pp in random batches; extreme-BCS contrasts reach 25–31 pp).

Conclusion. Training-free LLM extraction degrades along two separable dimensions—count-associated recall decline and confusability-associated type substitution—replicated on four models (8B–70B); decomposition fails via merge amplification and routing failure.

Limitations

1. **Model coverage.** Primary experiments use a single model (Qwen3-32B); cross-model validation covers the core extreme-BCS setting but does not replicate all secondary analyses (error taxonomy, prompt sensitivity, TOST equivalence tests). Whether these secondary findings generalize beyond Qwen3-32B remains open; see Appendix U and Table 4 for detailed model-wise results.

2. **Effect size and statistical design.** Pairwise SCS explains a small fraction of type confusion variance; entity frequency, prompt position, and mention ambiguity likely account for the remainder. The count $\times$ BCS interaction is bounded on FewRel but nonzero on DocRED, partially confounded by BCS range restriction at higher counts (Appendix C); logit-scale evidence remains inconclusive. DocRED count curves use 5 batches per level due to higher per-batch evaluation cost. BCS-controlled experiments use a single seed with greedy decoding; residual variance is small (Appendix R) but precise magnitudes may vary.

3. **Task scope.** Format independence is confirmed via paired constrained/unconstrained comparison (Appendix K). BCS batch-level prediction is task-dependent: significant in RE but null in NER, where errors scatter across miss, boundary, and type modes rather than concentrating in type substitution. A post-hoc relaxed-matching evaluation confirms that relaxing the span criterion does not recover the BCS signal (Appendix Y), ruling out evaluation stringency as the explanation.

4. **Causal mechanism.** The study establishes correlational evidence with controlled stratification; mechanistic analysis at the attention or logit level is needed to confirm the causal pathway. A post-hoc renaming experiment (Appendix X) suggests that SCS may proxy deeper conceptual overlap rather than surface label similarity; however, this remains post-hoc and requires more controlled causal analysis.

5. **Data contamination.** FewRel (2018) may appear in pretraining corpora, but train-origin relations show numerically lower accuracy than val-origin, opposite to the contamination hypothesis. DocRED (2019) provides a lower-contamination-risk replication with consistent patterns (Appendix Q).

6. **Domain and language scope.** All evaluations use English-language datasets derived from Wikipedia or knowledge bases (Wikidata, FrameNet). Whether confusability effects hold for domain-specific schemas (e.g., biomedical NER, legal RE) or languages with different naming conventions is untested. Sample sizes (1,000 instances per dataset) are sufficient for our primary effects but the pairwise SCS correlation approaches the detection floor.

Multiple comparisons. Cross-dataset comparisons serve as independent replications and do not require family-wise correction (Rothman, 1990) (noting that within-study prompt comparisons involve correlated observations rather than independent tests). Within-dataset secondary analyses involve multiple looks; we report individual p-values and note that core claims rest on $p < 0.001$ (Dror et al., 2018; Card et al., 2020).

Responsible NLP Research Checklist

For each question, the answer is **Yes**, **No**, or **N/A**. If Yes, the relevant section is indicated. If No, a justification is provided.

A For Every Submission

A1. Did you describe the limitations of your work?

Yes. The Limitations section (after §6) discusses model coverage, effect size and statistical design (including TOST equivalence), task scope, causal mechanism, data contamination, and domain/language scope.

A2. Did you discuss any potential risks of your work?

Yes (Limitations). SCS/BCS may be misused as a standalone deployment criterion. Since they are proxies for confusability and do not capture domain-specific ambiguity or downstream harms, they should be used for diagnostic monitoring rather than automatic schema rejection or high-stakes deployment decisions.

A3. Do the abstract and introduction summarize the paper's main claims?

Yes. The abstract and §1 state the two-dimensional degradation pattern (count and

confusability), the cross-model consistency check, and the negative SDGD result.

A4. Did you use AI assistants (e.g., ChatGPT, Copilot) when working on this paper?

Yes. We used Claude (Anthropic) for writing assistance and experimental code development. All scientific claims, experimental designs, and analyses were conducted by the authors.

B Scientific Artifacts

B1. Did you cite the creators of artifacts you used?

Yes. Few-NERD (Ding et al., 2021), FewRel (Han et al., 2018), DocRED (Yao et al., 2019), MAVEN (Wang et al., 2020b), Qwen3-32B (Qwen Team, 2025), Meta-Llama-3.1-70B-Instruct and Meta-Llama-3.1-8B-Instruct (Dubey et al., 2024), Gemini 2.5 Flash (Gemini Team, 2025), vLLM (Kwon et al., 2023), and all-MiniLM-L6-v2 (Wang et al., 2020a) are cited in §3.1 and §3.5.

B2. Did you discuss the license or terms for use and/or distribution of any artifacts?

Yes. All four datasets are publicly available research benchmarks. Few-NERD is released under CC-BY-SA 4.0; FewRel, DocRED, and MAVEN are publicly released for research use. Open-weight models are used under their respective licenses; Gemini 2.5 Flash is accessed via the Google API under its terms of service.

B3. Did you discuss if your use of existing artifacts is consistent with their intended use?

Yes. All datasets are used for their intended purpose (IE evaluation). No data is used outside the scope specified by the dataset creators.

B4. Did you discuss whether the data contains personally identifiable information or offensive content?

N/A. We use established benchmarks that have been vetted by the research community. No new data collection or annotation was performed.

B5. Did you provide documentation of the artifacts, e.g., coverage of languages, coverage of domains, etc.?

Yes. Table 1 reports type counts, evaluation unit counts, domain characteristics, and BCS ranges for each dataset. §3.1 describes dataset provenance and sampling procedures.

B6. Did you report relevant statistics for your dataset(s)?

Yes. §3.1 and Table 1 report evaluation set sizes, type counts, and sampling methodology. Count levels and batch sizes are specified in §3.3.

C Computational Experiments

C1. Did you report the number of parameters in the models used, the total computational budget, and computing infrastructure?

Yes. §3.5: Qwen3-32B (32B parameters, FP8), Meta-Llama-3.1-70B-Instruct (70B parameters, FP8), and Meta-Llama-3.1-8B-Instruct (8B parameters). Gemini 2.5 Flash was additionally accessed via API for cross-architecture validation. Inference was conducted on NVIDIA RTX PRO 6000 GPUs (96 GB) served via vLLM. Total compute was approximately 72 GPU-hours of inference (no training).

C2. Did you discuss the experimental setup, including hyperparameter search and best-found values?

Yes. §3.3 describes both experimental paradigms (count curves and BCS-controlled). §3.5 reports decoding parameters (greedy, temperature $= 0$, seed $= 42$) and prompt versions. No hyperparameter search was conducted as this is a zero-shot evaluation study.

C3. Did you report descriptive statistics about your results?

Yes. All key metrics include 95% bootstrap confidence intervals (10,000 resamples, percentile method) as stated in §3.5. Spearman correlations are reported with p-values throughout §4.1–4.5. Permutation tests are used for slope comparisons.

C4. Did you report implementation details, e.g., versions of packages?

Yes. §3.5 specifies vLLM v0.20.2 for model serving. Embedding models are named with version identifiers (§3.2). Code and scripts will be released upon acceptance.

D Human Annotators or Crowdworkers

D1. Did you report the full text of instructions given to participants?

N/A. No human annotation or crowdsourcing was conducted.

D2. Did you report information about how you recruited and paid participants?

N/A.

D3. Did you discuss whether and how consent was obtained?

N/A.

D4. Did you report whether the data collection was approved by an institutional review board (IRB)?

N/A.

D5. Did you report the basic demographic and geographic characteristics of the annotator population?

N/A.

E AI-Assisted Research

E1. Did you report the use of AI assistants?

Yes. See A4 above. AI assistants were used for code development and document formatting. The research design, execution, and interpretation are the authors' own work.

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A Additional Figures

B SDGD Full Protocol and Results

We construct a thresholded confusability graph $G = (V, E)$ where each type is a vertex and an edge connects types t_i, t_j whenever $\mathrm{SCS}(t_i, t_j) > \tau$. Greedy graph coloring (LARGEST_FIRST) assigns each type to a color such that no two adjacent (i.e., highly confusable) types share a color. Each color class forms one extraction batch, and predictions are merged via set union. We compare four modes: (i) **single_pass**, the undecomposed baseline; (ii) **sdgd_zero**, SCS-guided partitioning; (iii) **random_batch**, random partitioning

Count	Llama-3.1-8B	Qwen3-32B
1	0.702	0.640
64	0.273	0.304
Decline	−43 pp	−34 pp

Table 3: Cross-model count-curve replication on Few-NERD (text-match recall). Both models show a general declining trend in recall with schema size. Per-batch correlation $r = 0.725$. Qwen3-32B values use the v2 prompt (§3.5) for fair cross-model comparison; the count-curve results in §4.1 use the v1 prompt, yielding higher absolute recall (0.72 vs. 0.64 at count $= 1$) due to over-prediction under v1.

	Llama-3.1-8B			Qwen3-32B		
Count	Low	High	Δ	Low	High	Δ
8	78.1%	51.0%	−27.1	82.3%	51.0%	−31.2
16	75.5%	51.6%	−23.9	82.8%	51.6%	−31.2

Table 4: Cross-model BCS-controlled FewRel accuracy. High-BCS accuracy converges to ≈51% regardless of model, while low-BCS accuracy reflects model capacity differences. The confusability gap (Δ, in pp) is 77–87% as large on Llama-3.1-8B as on Qwen3-32B.

		BCS Bin						
Task	Error	Solo	Low	Mid	High	V.High	ρ	p
NER	IntraSub	0%	1.6%	10.6%	49.4%	—	1.000	<0.001
RE	IntraSub	0%	4.2%	13.0%	20.3%	32.3%	0.954	<0.001
DocRE	IntraSub	—	4.8%	22.4%	37.0%	41.9%	0.843	<0.001
NER	Missed	—	42.8%	64.8%	49.9%	—	—	—
RE	Missed	—	10.4%	11.3%	13.1%	15.7%	—	—
DocRE	Missed	—	28.1%	22.6%	21.6%	16.2%	—	—

Table 5: Intra-batch substitution (IntraSub) and missed-entity rates by BCS bin. Spearman ρ computed over $n = 148$ batches (NER), $n = 154$ (RE), and $n = 110$ (DocRE). IntraSub scales monotonically with BCS across all three datasets. DocRE missed rates decrease with BCS, indicating high-confusability errors shift from omission to substitution.

Mode	# Batches	Precision	Recall	F1
single_pass	1	0.563	0.368	0.445
sdgd_zero (v1)	8	0.210	0.314	0.251
random_batch (v1)	8	0.249	0.446	0.320
per_type	66	0.194	0.586	0.292
sdgd_zero (v2)	4	0.324	0.191	0.241
random_batch (v2)	8	0.331	0.276	0.301

Table 6: Few-NERD extraction under decomposition modes. V1 uses union merge; v2 uses confidence-gated merge with `min_batch_size=8` (reducing 8→4 batches for sdgd_zero). All decomposition modes produce lower F1 than the single-pass baseline. All modes use the v1 prompt.

with matched batch sizes; and (iv) **per_type**, the extreme case where each type is extracted in isolation.

Single-pass extraction achieves F1 = 0.445 (P = 0.563, R = 0.368). SDGD obtains F1 = 0.251 (P = 0.210, R = 0.314), a 44% relative drop. Random batching reaches F1 = 0.320 (P = 0.249, R = 0.446), and per-type extraction yields F1 = 0.292 (P = 0.194, R = 0.586). Every decomposition mode falls below the single-pass baseline (Table 6).

Decomposition boosts recall (per-type recall reaches 0.586 versus single-pass 0.368) but devastates precision (0.194 versus 0.563). When the same text is presented to the model k times with different type subsets, each pass independently generates entity predictions. Because LLM outputs are not calibrated across passes, the union merge step accumulates overlapping and contradictory predictions and inflates false positives.

Our v2 merge ranks competing predictions by the LLM's self-reported confidence and retains only the highest-scoring candidate for each span. Confidence gating improves precision (sdgd_zero: 0.210→0.324; random_batch: 0.249→0.331), but at the cost of disproportionate recall loss (0.314→0.191; 0.446→0.276), and both modes produce lower F1 (sdgd_zero: 0.251→0.241; random_batch: 0.320→0.301).

B.1 Practical Routing Strategies on RE

We evaluate five non-oracle routing strategies on FewRel 80-relation RE using graph-coloring sub-schema decomposition (same grouping as Table 10). Each strategy selects a single sub-schema for each test instance and returns that sub-schema's prediction.

C Factorial Analysis and Separability Test

To formally test whether count and confusability effects are additive, we fit a multiple regression (accuracy $\sim \log_2$ count + BCS + count×BCS) across 69 batches from FewRel spanning five count levels ($k \in \{4, 8, 16, 32, 40\}$). Both main effects are significant (count: $p = 0.008$; BCS: $p = 0.0002$; $R^2 = 0.52$), but the interaction term is negligible (0.47 pp). A TOST equivalence test with a ±3 pp margin confirms that the interaction is bounded well below practical significance ($p_{\text{TOST}} = 0.003$); this supports bounded interaction between count and confusability effects.

A 2 × 2 factorial sensitivity check on FewRel

Strategy	Accuracy	Δ vs. Single-Pass
Oracle (gold label)	72.4%	+22.0 pp
Max-conf (positive only)	36.5%	−14.0 pp
None-aware	36.5%	−14.0 pp
Margin (positive only)	35.9%	−14.5 pp
Max-conf (all)	27.7%	−22.7 pp
Random	14.7%	−35.7 pp
Single-pass (no decomposition)	50.4%	—

Table 7: Routing strategy comparison on FewRel 80-relation RE with graph-coloring sub-schema decomposition. Oracle requires gold labels. All practical (non-oracle) strategies perform below the single-pass baseline. The failure stems from constrained decoding (enum constraint), which forces confident predictions in every sub-schema: 73.4% of instances receive positive predictions from 3–5 sub-schemas simultaneously, with per-sub-schema none-rates of only 20–40%.

Margin (pp)	t_{upper}	p_{TOST}	Conclusion
±1	−0.59	0.278	Inconclusive
±2	−1.70	0.047	Equivalent
±3	−2.81	0.003	Equivalent
±5	−5.02	<0.001	Equivalent

Table 8: TOST sensitivity analysis across equivalence margins. The interaction effect (0.47 pp, 90% CI [−1.0, 2.0] pp) is bounded below 2 pp at marginal significance ($p = 0.047$); the ±3 pp choice does not drive the equivalence conclusion. At ±1 pp, equivalence is inconclusive ($p = 0.278$), meaning interaction could reach approximately half the main effect magnitude.

(count $\in \{8, 16\}$, BCS low/high; $n = 54$) was inconclusive ($p_{\text{TOST}} = 0.56$) due to limited power; this does not contradict the primary finding but limits its generalization beyond the regression framework. The Few-NERD factorial (§4.4) similarly did not reach significance ($p = 0.50$). The regression approach provides the stronger test by leveraging the full count range.

Equivalence margin justification. We set the ±3 pp margin based on three converging anchors. *Empirical*: typical random-batch confusability variation produces 2–5 pp accuracy differences (batch SD ≈ 4.6 pp at count 32). *Practical*: a <3 pp accuracy difference falls below deployment-relevant thresholds. *Statistical*: 3 pp ≈ 0.5× batch SD, equivalent to Cohen's $d \approx 0.5$ (medium effect threshold; Lakens 2017). A 3 pp interaction atop the 2.2 pp confusability main effect would represent >130% amplification, fundamentally changing whether confusability warrants mitigation. At ±2 pp (marginally equivalent, $p = 0.047$), interaction remains bounded below the main effect itself.

Logit-scale sensitivity. Because accuracy near 50% compresses interaction effects on the linear scale, we refit the regression on logit-transformed accuracy. The logit-scale interaction (0.111 logit units, $p = 0.047$, 90% CI [0.019, 0.203]) is marginally significant, and TOST at the equivalent ±3 pp margin (±0.156 logit at median accuracy) is inconclusive ($p_{\text{TOST}} = 0.209$). However, back-transforming the interaction to the probability scale at the observed floor ($p = 0.51$) yields 2.78 pp, still within the ±3 pp margin; at the median baseline ($p = 0.65$), the back-transformed effect is 2.53 pp. The logit TOST is inconclusive due to limited power ($n = 69$, wide CI) rather than a large effect size: the point estimate remains below the equivalence bound at all observed baselines. We retain the linear-scale TOST as the primary test and report this logit analysis as a sensitivity check.

DocRED factorial (expanded). With expanded batches (n=55 per count level, up from n=5), the DocRED 2 × 2 factorial (count $\in \{16, 32\}$, BCS median split) yields an apparent interaction (−9.4 pp, 95% CI [−12.9, −6.0]): the BCS effect is −25.0 pp at count 16 but −15.7 pp at count 32. However, this attenuation is confounded by BCS range restriction: higher-count batches have narrower BCS ranges (spread 0.26 vs. 0.38), limiting achievable extremes.

Range-matched robustness check. To control for the BCS range confound, we restrict both count levels to the overlapping BCS range [0.43, 0.70]. This retains 29 of 55 batches at count 16 (all 55 at count 32 fall within the range). Within this matched range, the BCS effect at count 16 shrinks from 15.95 pp to 9.37 pp (reflecting the narrower BCS span), while count 32 remains at 8.44 pp. The interaction drops from 7.52 pp (95% CI [1.49, 12.83]) to 0.93 pp (95% CI [−7.13, 7.44])—the CI spans zero, consistent with no interaction. Spearman correlations remain comparable ($\rho = -0.65$ at count 16, $\rho = -0.69$ at count 32), confirming that the underlying BCS–accuracy relationship is robust across counts while the raw interaction magnitude was an artifact of differing achievable BCS ranges. This result aligns with the FewRel TOST equivalence finding and supports bounded association as the cross-dataset pattern.

D Within-Pool CI Details

The quartile Δ for 80-pool 16-relation subsets (-3.92 pp, 95% CI [-8.17, $+0.38$]) crosses zero, reflecting the compressed BCS range within that pool (0.46–0.68 vs. 0.48–0.71 for the independent pool). Within this narrow band, the quartile contrast lacks statistical power to detect the effect that the regression identifies. The full 80-relation condition ($n = 25$) is underpowered for both the slope (95% CI [-1.303, $+1.695$]) and quartile analyses; we include it in Table 16 for completeness but do not draw inferential conclusions from it.

E Controlling for Centrality Confounds in MAVEN Per-Type Analysis

Section 4.5 identifies a centrality confound in MAVEN's per-type BCS–F1 correlation: both embeddings show a *positive* raw correlation, opposite to the expected direction. Here we formalize this analysis by identifying specific confounders for each embedding and testing whether the BCS–F1 relationship survives partial correlation controlling for these confounders.

Confound identification. For each embedding, we computed the correlation between mean BCS and candidate confounders (type frequency, degree centrality at threshold 0.7) across MAVEN's 149 active event types. The dominant confound differs by embedding:

- **MiniLM**: BCS correlates with log type frequency ($r = 0.201$). Frequent types have more training signal and are extracted more reliably, which produces the positive BCS–F1 association.
- **Qwen3**: BCS correlates strongly with degree centrality ($r = 0.854$). In the Qwen3 embedding space, high-BCS types are those with many semantic neighbors, central, well-represented types that the model extracts well.

Partial correlation results. Table 9 reports raw Spearman ρ and partial Pearson r after controlling for the identified confounders. Under single-variable control, both correlations weaken to non-significance. Under joint control (log frequency + degree centrality simultaneously), neither embedding retains a significant BCS–F1 association.

Interpretation. The two embeddings encode different aspects of the same underlying confound. MiniLM, a smaller sentence encoder, produces a tighter embedding cluster for MAVEN's FrameNet-derived types; here, type frequency is the primary confounder because frequent types accumulate higher mean similarity to the dense cluster. Qwen3, with a larger representation space, differentiates types enough that frequency is not the dominant factor, but degree centrality (the count of strong semantic neighbors) becomes the confound instead. In both cases, the positive raw correlation reflects a centrality artifact: types that are semantically central are both high-BCS and easy to extract.

Under joint control, both embeddings yield $p > 0.2$; the raw positive BCS–F1 correlation on MAVEN is thus fully attributable to centrality-related confounders. This does not invalidate the confusability hypothesis; the pairwise SCS–confusion correlation on MAVEN remains significant (MiniLM $r = 0.129$, $p < 0.001$; Table 2). Rather, it demonstrates that per-type aggregation (mean BCS) conflates confusability with centrality in semantically homogeneous schemas, and that pairwise analysis or explicit confound adjustment is required in such settings.

Embedding	Control	Raw ρ	Raw p	Partial r	Partial p
MiniLM	—	+0.208	0.011	—	—
MiniLM	log freq	—	—	+0.133	0.11
MiniLM	log freq + degree	—	—	—	0.23
Qwen3	—	+0.184	0.025	—	—
Qwen3	degree centrality	—	—	+0.063	0.44
Qwen3	log freq + degree	—	—	—	0.44

Table 9: MAVEN per-type BCS–F1 correlation before and after controlling for centrality confounders. Raw correlations are Spearman ρ; partial correlations are Pearson r from residualized regression. Degree centrality is computed on the SCS graph at threshold 0.7. All values from experiment `exp_maven_residualized_bcs`.

F Batch Composition Results

Condition	Accuracy	Mean BCS
Single-pass (80 types)	0.504	0.557
Random batching (15 seeds)†	0.788 ± 0.018	0.559
Confus.-aware (graph coloring)	**0.802**	0.552

Table 10: Batch composition on FewRel 80-relation RE with oracle routing. †Mean ± SD across 15 random seeds.

See the accompanying discussion in §5.

G Prompt Templates

This appendix reproduces the complete prompt templates used for all three extraction tasks. All prompts are injected into a single `user` message (no system prompt); the model's chat template adds the system/assistant framing. Both versions provide a JSON-formatted type schema to the vLLM constrained-decoding engine; the schema is *not* included in the natural-language prompt itself. We use greedy decoding (`temperature=0`, `seed=42`) throughout.

Notation. Placeholders are shown in `{braces}`. Text **added in v2** (relative to v1) is marked with ▷. Text **removed in v2** is marked with ~~strikethrough~~.

G.1 Named Entity Recognition (Few-NERD)

```
Extract all named entities from the
    following text.
Only extract entities of the following
    types:
- {type_1}: {definition_1}
- {type_2}: {definition_2}
  ...

Text: {input_text}

Output as JSON: [{"text": "...",
    "type": "...",
              "start": ..., "end": ...}]
```

v1 constrained-decoding schema:

```
{"type": "array",
 "items": {"type": "object",
   "properties": {
     "text": {"type": "string"},
     "type": {"type": "string",
             "enum": [<sorted_types>]},
     "start": {"type": "integer"},
     "end": {"type": "integer"}},
   "required":
      ["text","type","start","end"]}}
```

```
Extract all named entities from the
    following text.
Only extract entities of the following
    types:
- {type_1}: {definition_1}
- {type_2}: {definition_2}
  ...

Text: {input_text}
```

▷ Added in v2:

```
Only extract entities that clearly
    match one of the
given types. If no entities of these
    types exist in
the text, output an empty list [].

Output as JSON: [{"text": "...",
    "type": "..."}]
```

v2 constrained-decoding schema:

```
{"type": "array",
 "items": {"type": "object",
   "properties": {
     "text": {"type": "string"},
     "type": {"type": "string",
             "enum": [<sorted_types>]}},
   "required": ["text","type"]}}
```

v1 → v2 differences (NER).

1. **Offset fields removed.** The `start`/`end` integer fields are dropped from both the natural-language instruction and the JSON schema. In v1, 96.5% of predicted offsets were hallucinated (§3.5).
2. **Format constraints added.** Two sentences are appended: (a) "Only extract entities that clearly match one of the given types" suppresses spurious predictions; (b) "If no entities … output an empty list `[]`" provides an explicit empty-output path.

G.2 Event Detection (MAVEN)

```
Extract all event triggers from the
    following text.
A trigger is the word or phrase that
    most clearly
expresses the event occurrence. Only
    extract triggers
of the following event types:
- {type_1}: {definition_1}
- {type_2}: {definition_2}
  ...

Text: {input_text}

Output as JSON: [{"text": "...",
    "type": "...",
              "start": ..., "end": ...}]
```

```
Extract all event triggers from the
    following text.
A trigger is the word or phrase that
    most clearly
expresses the event occurrence. Only
    extract triggers
of the following event types:
- {type_1}: {definition_1}
- {type_2}: {definition_2}
  ...

Text: {input_text}
```

▷ Added in v2:

```
Only extract triggers that clearly
    match one of the
given event types. If no events of
    these types exist
in the text, output an empty list [].
```

```
Output as JSON: [{"text": "...",
    "type": "..."}]
```

The JSON schemas for ED are identical to the NER schemas above; only the natural-language framing differs ("event triggers" instead of "named entities").

v1 → v2 differences (ED). Same two changes as NER: offset fields removed, format constraints added.

G.3 Relation Extraction (FewRel)

RE uses only a v2-style prompt (no offset fields are applicable to classification).

```
Given the following sentence and two
    entities,
classify the relation between them.

Sentence: "{sentence}"
Head entity: "{head_entity}"
Tail entity: "{tail_entity}"

Candidate relations: {type_1},
    {type_2}, ...

Output a single JSON object:
  {"relation": "<relation_type>"}
If none of the candidate relations
    apply, output:
  {"relation": "none"}
```

Constrained-decoding schema:

```
{"type": "object",
 "properties": {
   "relation": {"type": "string",
     "enum": [<sorted_types>, "none"]}},
 "required": ["relation"]}
```

G.4 Type Definitions

All prompts include per-type natural-language definitions sourced from the respective dataset documentation:

- **Few-NERD:** 66 fine-grained entity types with definitions derived from the Few-NERD type ontology (Ding et al., 2021).
- **MAVEN:** 168 event types with FrameNet-based definitions (Wang et al., 2020b).
- **FewRel:** 80 relation types with Wikidata property descriptions (Han et al., 2018).

Type definitions are injected as "– `{type_name}: {definition}`" lines in alphabetical order. The number of types per prompt varies by experimental condition (count = 1 to full schema; §3.3).

H Prompt Version Sensitivity Analysis

We assess whether the confusability effects reported in §4.3–4.4 are robust to prompt design by replicating key experiments under the v2 prompt (§3.5; full templates in Appendix G).

RE (FewRel). Under the v1 prompt, the extreme-batch BCS gap (Table 13) is substantial. Under v2, with n=25 batches per group and wider BCS contrast (low-BCS range [0.356, 0.367]; high-BCS range [0.791, 0.811]), the gap is -25.5 pp (low-BCS 85.2% vs. high-BCS 59.7%; 95% CI [22.4, 28.6]; $p < 0.001$; Pearson $r = -0.919$). The effect is significant, directionally consistent with v1, and only modestly attenuated in magnitude (-5.7 pp reduction). An earlier n=5 evaluation with narrower BCS contrast (0.21) yielded a smaller gap of -7.9 pp (95% CI [-5.2, $+21.5$]), illustrating how limited statistical power and restricted BCS range can obscure the underlying signal, a cautionary note for small-sample prompt sensitivity studies.

NER (Few-NERD). The batch-level BCS–F1 correlation is null under both prompt versions. Pooling across three seeds and all count levels ($n = 450$), the correlation is $r = 0.004$ ($p = 0.94$). An earlier small-sample analysis ($n = 27$, single seed) had reported positive correlations ($r = +0.50/+0.58$), but these were driven by an accidental BCS–type-frequency confound: those 27 batches exhibited $r(\text{BCS}, \text{freq}) = 0.53$, creating a spurious path through frequency to F1. After controlling for type frequency, the partial correlation is non-significant in both the small ($r = 0.28$, $p = 0.16$) and large ($r = -0.12$, $p = 0.29$) samples. The pairwise SCS–confusion analysis (§4.2), which is immune to this batch-level aggregation artifact, remains the primary evidence for the NER confusability dimension.

Interpretation. The n=25 v2 result confirms that the confusability effect is directionally consistent across prompt designs, with attenuated magnitude: the gap narrows (-5.7 pp) but remains highly significant ($p < 0.001$, $r = -0.919$). The earlier n=5 non-significance reflected insufficient power and narrow BCS contrast, not absence of the effect.

V2's stricter format constraints raise the baseline accuracy (low-BCS 85.2% vs. 82.3%), but type confusion under high confusability persists (59.7% vs. 51.0%). Prompt design modulates the practical magnitude of the confusability effect but does not eliminate it, ruling out prompt-specific confounding as the primary source. The underlying confusability signal (pairwise SCS–confusion correlation and the directional BCS–accuracy relationship) persists across both prompt versions.

H.1 Cross-Prompt-Variant BCS Correlation

To verify that the BCS–accuracy relationship is not an artifact of a specific prompt template, we evaluate four structurally distinct prompt variants on FewRel 80-relation RE at two count levels (8 and 32), with 10 random batches per condition.

Variant	Count	Mean Acc	r	p
V2 (definition-first)	8	0.769	-0.333	0.348
V2 (definition-first)	32	0.564	-0.648	0.043
Instruction-rewrite	8	0.847	-0.223	0.536
Instruction-rewrite	32	0.662	-0.469	0.171
Chain-of-thought	8	0.749	$+0.254$	0.478
Chain-of-thought	32	0.617	-0.566	0.088
Minimal	8	0.852	-0.170	0.638
Minimal	32	0.615	-0.771	0.009

Table 11: BCS–accuracy Pearson correlation across four prompt variants. At count 32, all four variants show negative BCS–accuracy correlations ($r \leq -0.47$), with two reaching $p < 0.05$. At count 8, the narrower BCS range reduces statistical power.

At count 32, all four variants produce negative BCS–accuracy correlations (r ranges from -0.47 to -0.77), with V2 ($p = 0.043$) and Minimal ($p = 0.009$) reaching significance and Chain-of-thought approaching it ($p = 0.088$). At count 8, correlations are weaker and nonsignificant across all variants, consistent with the narrower BCS range at lower schema complexity reducing the detectable effect size. Rather than applying Bonferroni correction, which assumes independent hypotheses inappropriate for correlated prompt variants testing the same underlying signal, we note directional consistency: 7 of 8 conditions show negative BCS–accuracy slopes, consistent with the primary finding (Rothman, 1990). These results confirm that the confusability effect arises from schema composition, not prompt-specific phrasing.

I BCS Aggregation Sensitivity

We define Batch Confusability Score (BCS) as the *mean* pairwise SCS across all type pairs in a batch (§4.3). An alternative definition takes the *maximum* pairwise SCS within each batch to capture the single most confusable pair. Here we compare both aggregation methods to verify that core conclusions are robust to this choice.

Aggregation	Condition	Slope	95% CI	r	p
MEAN (primary)	16-rel	-0.509	$[-0.686, -0.336]$	-0.467	< 0.001
MEAN	80-pool 16-sub	-0.401	$[-0.701, -0.069]$	-0.221	0.027
MAX	16-rel	-0.449	$[-0.552, -0.346]$	-0.637	< 0.001
MAX	80-pool 16-sub	-0.401	$[-0.701, -0.069]$	-0.221	0.027

Table 12: BCS–accuracy regression under mean vs. max pairwise SCS aggregation (FewRel, 16-relation batches). Permutation test for slope equivalence across pools: MEAN $p \approx 0.60$; MAX $p \approx 0.44$. The 80-pool row is identical under both definitions because these batches were collected using mean-aggregated BCS; the MAX column reflects the same data for completeness.

MAX aggregation yields stronger correlations ($r = -0.637$ vs. -0.467 for 16-rel), consistent with the hypothesis that the single most confusable pair in a batch predicts substitution errors. Both definitions produce significant negative slopes across all conditions, and the permutation tests confirm no significant difference between independent and sampled pools under either definition. We adopt MEAN as the primary definition because it better represents average pairwise difficulty across all schema combinations, while MAX may overweight a single outlier pair. The choice does not affect any paper conclusion.

J BCS-Controlled Detailed Results

Schema	Count	Low-BCS	High-BCS	Δ (pp)	95% CI
16-rel	8	88.9%	79.8%	-9.1	—
80-rel	8	82.3%	51.0%	-31.2	[17.7, 42.7]
80-rel	16	82.8%	51.6%	-31.2	[22.9, 39.6]

Table 13: BCS-controlled FewRel accuracy. Low-BCS batches contain maximally dissimilar types; high-BCS batches contain maximally similar types. The 80-relation extreme-batch gap reflects the wider BCS range; regression slopes are consistent across pools ($p = 0.60$; Table 16; see §4.3 for typical-batch magnitudes). The 16-relation condition used a single extreme batch per BCS level; bootstrap CIs require multiple batches.

Adjusted R^2 and confidence intervals. DoCRED regressions (§4.3) use OLS on $n = 55$ batches (15 high-BCS + 15 low-BCS + 25 random) per count level, with one predictor. At count 16, adjusted $R^2 = 0.646$ (unadjusted: 0.653; 95% CI [0.53, 0.77]); at count 32, adjusted $R^2 = 0.606$ (unadjusted: 0.614; 95% CI [0.47, 0.75]). The small

shrinkage ($\Delta R^2 \leq 0.01$) indicates the unadjusted values are not inflated by the predictor-to-sample ratio.

K Format Independence: Paired Constrained vs. Unconstrained Comparison

To test whether the BCS–accuracy relationship depends on the output format, we run both constrained (enum-based) and unconstrained (free-text) decoding on the same 25 batch type sets from the FewRel 80-relation pool (13 low-BCS, 12 high-BCS; count 8). Each batch uses identical instances and type sets; only the decoding format differs.

Metric	Constrained	Unconstrained
Mean accuracy	79.5%	79.6%
Pearson r (BCS vs. acc)	-0.448 ($p = 0.025$)	-0.439 ($p = 0.028$)
Spearman ρ (BCS vs. acc)	-0.157 ($p = 0.455$)	-0.144 ($p = 0.492$)
Paired Fisher z-test for correlation equality		
Pearson z	-0.04 ($p = 0.97$)	
Spearman z	-0.04 ($p = 0.97$)	

Table 14: Paired comparison of constrained vs. unconstrained decoding on 25 identical FewRel batch type sets. Both formats yield virtually identical BCS–accuracy correlations (Fisher $z \approx 0$), confirming the confusability effect is format-independent. Mean accuracy is also unchanged ($\Delta = -0.08$ pp), ruling out a format-driven accuracy shift.

The near-zero Fisher z statistics ($|z| < 0.05$, $p > 0.96$) confirm that the BCS–accuracy correlation is indistinguishable between formats. Both Pearson r values are significant ($p < 0.03$), reflecting the bimodal BCS distribution of extreme batches (13 low + 12 high); the monotonic Spearman ρ is modest and non-significant under both formats, consistent with the typical-batch BCS effect size ($\sim$2 pp; §4.3). This result establishes that the confusability-associated degradation pattern is robust across output formats, not an artifact of forced-choice decoding.

Per-batch paired results. Table 15 reports per-batch accuracy under both formats. Of 25 batches, 23 yield *identical* accuracy; the two exceptions (high_5, high_11) differ by a single sample (1/96 $\approx$ 1.04 pp). Only 2 out-of-vocabulary predictions occur in unconstrained decoding (OOV rate 0.08%), confirming that the model generates valid type labels even without format constraints.

Batch	Group	BCS	Constr.	Unconstr.	Δ
low_0	L	0.472	83.3	83.3	0.0
low_1	L	0.647	81.3	81.3	0.0
low_2	L	0.665	87.5	87.5	0.0
low_3	L	0.672	82.3	82.3	0.0
low_4	L	0.674	67.7	67.7	0.0
low_5	L	0.678	72.9	72.9	0.0
low_6	L	0.679	80.2	80.2	0.0
low_7	L	0.687	87.5	87.5	0.0
low_8	L	0.692	77.1	77.1	0.0
low_9	L	0.692	91.7	91.7	0.0
low_10	L	0.692	87.5	87.5	0.0
low_11	L	0.709	84.4	84.4	0.0
low_12	L	0.709	68.8	68.8	0.0
high_0	H	0.712	91.7	91.7	0.0
high_1	H	0.724	77.1	77.1	0.0
high_2	H	0.735	67.7	67.7	0.0
high_3	H	0.742	86.5	86.5	0.0
high_4	H	0.742	86.5	86.5	0.0
high_5	H	0.747	81.3	82.3	−1.0
high_6	H	0.768	83.3	83.3	0.0
high_7	H	0.778	77.1	77.1	0.0
high_8	H	0.791	71.9	71.9	0.0
high_9	H	0.820	70.8	70.8	0.0
high_10	H	0.823	92.7	92.7	0.0
high_11	H	0.971	49.0	50.0	−1.0

Table 15: Per-batch accuracy (%) under constrained and unconstrained decoding on 25 identical FewRel batch type sets (count 8, 96 instances each). 23/25 batches yield identical accuracy; the two exceptions differ by exactly one sample ($\Delta = -1.04$ pp). L = low-BCS, H = high-BCS.

L Within-Pool Analysis Methodology

A potential confound in the cross-scale comparison is that the 16-relation and 80-relation pools contain different types, so the observed gap could reflect pool composition rather than confusability. To control for this, we draw 100 random 16-relation subsets from the 80-relation pool and evaluate under the same BCS-controlled protocol.

	Regression (BCS → Accuracy)				Quartile Δ	
Setting	Slope	95% CI	r	p	Δ (pp)	95% CI
Independent 16-rel (n=100)	−0.509	[−0.686, −0.336]	−0.467	< 0.001	−5.31	[−7.44, −3.19]
80-pool → 16-sub (n=100)	−0.401	[−0.701, −0.069]	−0.221	0.027	−3.92	[−8.17, +0.38]
Full 80-rel (n=25)†	+0.221	[−1.303, +1.695]	+0.064	0.76	−1.04	[−10.12, +8.04]

Permutation test (16-rel vs. 80-pool slopes): $p \approx 0.60$ (5,000 permutations)
†BCS range (IQR = 0.033) insufficient for regression; slope not interpretable.

Table 16: Three-way BCS–accuracy analysis on FewRel. Slope: OLS of batch accuracy on BCS (bootstrap CIs, B=10,000). Δ: accuracy gap between top- and bottom-quartile BCS batches. See Appendix D for CI interpretation.

The within-pool slope (-0.401, CI excludes zero) is statistically indistinguishable from the independent 16-relation pool (-0.509; permutation $p \approx 0.60$). The smaller quartile gap (-3.92 pp vs. -5.31 pp) reflects the narrower BCS range within the 80-relation pool (Appendix D).

In the full 80-relation condition, natural BCS variation is insufficient (IQR $= 0.033$) to yield a detectable effect ($n = 25$, slope $= +0.221$, $p \approx 0.76$). This does not invalidate the confusability effect; the within-pool slope is significant ($p = 0.027$) when BCS is deliberately varied, but establishes that natural schema sampling at this scale does not produce sufficient BCS contrast for detection.

M SCS Embedding Ablation: Name-Only vs. Name+Definition

Table 17 compares predictive accuracy of name-only SCS against name+definition SCS across three datasets.

Dataset	SCS_{name} r	SCS_{def} r	Δr	Matrix corr.
Few-NERD	0.172	0.149	−0.023	0.894
MAVEN	0.129	0.164	+0.035	0.656
FewRel	0.178	0.193	+0.015	0.848

Table 17: Pearson correlation between SCS and empirical confusion rate. "Def" concatenates type name with definition before embedding. All correlations significant ($p < 0.001$). The inconsistent direction of Δr indicates that definitions provide no systematic benefit.

Few-NERD is the only dataset whose prompts include type definitions; for MAVEN and FewRel, prompts use type names alone, so name-only SCS directly matches model input. The high matrix correlation (0.66–0.89) confirms that type names capture the dominant semantic structure.

Held-out validation. From the 80-type confusion matrix, we extract the $\binom{16}{2} = 120$ pairs among the 16 validation-origin types (fully disjoint from the 64 training-pool types). The SCS–confusion Spearman correlation remains positive and significant ($\rho = 0.216$, $p = 0.018$), nearly identical to the full 80-type pool ($\rho = 0.218$, $p < 0.001$, $n = 3{,}160$ pairs), ruling out training-set contamination as the driver of the observed relationship.

N Pair-to-Batch Aggregation Simulation

The following simulation provides an illustrative explanatory model for how pair-level effects aggregate to batch-level variance. It is not the evidence base for our claims; the observed batch-level $R^2 = 0.218$ (§4.3) is a direct empirical measurement independent of this simulation.

Setup. We examine whether the pair-level $R^2 \approx 7\%$ (computed over non-zero confusion pairs) can aggregate to the observed batch-level $R^2 \approx 22\%$ via a Monte Carlo simulation. From the FewRel 16-relation experiment, we estimate the inter-pair SCS correlation for pairs sharing a common type ($\rho_{\text{SCS}} = 0.35$). For batches of $k = 8$ types ($\binom{8}{2} = 28$ pairs), we simulate 10,000 iterations: (1) generate 28 correlated pair-level effects with individual $R^2 = 0.074$; (2) average to obtain batch-level signals; (3) compute batch-level R^2.

Results. Simulated batch $R^2 = 0.219$ (95% CI [0.092, 0.361]), matching the observed value of 0.218. The effective amplification factor is $\sim 3\times$, explained by signal correlation within batches preserving variance while independent noise averages out (effective degrees of freedom ≈ 3.4, not 28).

Unconditional baseline. When the simulation uses the unconditional pairwise $r^2 = 0.013$ (computed over all type pairs including zero-confusion pairs; Table 2), the simulated batch-level R^2 drops to 0.044 (95% CI [0.001, 0.151]), well below the observed $R^2 = 0.218$. The analytical formula confirms: $\text{SNR} \times \text{EF}/(\text{SNR} \times \text{EF} + 1) = 0.042$ with the same effective factor $\text{EF} = 3.35$. This $\sim 5\times$ gap reflects the dilution from zero-confusion pairs: approximately 89% of type pairs never exhibit mutual confusion, contributing only noise to the unconditional correlation. The conditional model ($R^2 = 0.074 \rightarrow 0.219$) matches the observed value because it isolates the minority of semantically proximate pairs that actually compete during type resolution; BCS predicts batch accuracy through the density of *active* confusable pairs, not uniform contributions from all pairs.

Sensitivity to ρ. The inter-pair SCS correlation $\rho = 0.35$ is estimated from pairs sharing a common type. At $\rho = 0.20$ (halved correlation), a parallel simulation yields batch $R^2 \approx 0.22$, still matching the observed value; at $\rho = 0.50$, the correlation structure approaches the positive-definiteness boundary; $\rho = 0.35$ thus sits in a stable regime. The simulation is illustrative; the observed batch $R^2 = 0.218$ is the primary evidence.

Non-independence and ecological inference. Types within a batch share prompt context, so their extraction outcomes are not independent—a form of ecological correlation (Robinson, 1950). This non-independence is built into the design: BCS measures batch-level confusability precisely because co-present types interact during type resolution. The core batch-level $R^2 = 0.218$–0.578

(§4.3) is a direct empirical observation from regressing batch accuracy on BCS, not derived from the simulation above. The inter-pair SCS correlation $\rho = 0.35$ used in the simulation was estimated from the SCS correlation matrix independently of batch-level accuracy data.

O BCS Independence from Type Composition

FewRel's balanced design (700 instances per type) makes type frequency invariant across batches, ruling out frequency-driven confounding. To test whether BCS captures information beyond type composition effects, we decompose the explained variance in a joint model (accuracy $\sim$ BCS + mean type difficulty), where type difficulty is the per-type solo-baseline accuracy averaged across batch members. On 100 random 8-type batches from the 16-relation pool, the joint model explains $R^2 = 0.704$. Variance decomposition yields: BCS unique $R^2 = 0.259$ (36.8% of explained variance), composition unique $R^2 = 0.298$ (42.3%), and shared $R^2 = 0.147$ (20.9%). The near-orthogonality of predictors (VIF $= 1.06$, $r(\text{BCS}, \text{difficulty}) = -0.21$) confirms that BCS captures confusability-specific signal rather than a proxy for type difficulty.

P SCS vs. Simple Baselines

Dataset	Metric	Spearman ρ	p
FewRel (3,160 pairs)	SCS (embedding)	**0.218**	< 0.001
	Same-parent (WordNet)	0.097	< 0.001
	WordNet Wu-Palmer	0.073	< 0.001
	Levenshtein	0.016	0.363 (n.s.)
Few-NERD (1,953 pairs)	SCS (embedding)	**0.166**	< 0.001
	Levenshtein	0.152	< 0.001
	Same-parent (prefix)	0.146	< 0.001
	WordNet Wu-Palmer	0.054	0.070 (n.s.)†

Table 18: Pairwise confusion prediction: SCS vs. simple baselines. Spearman ρ between each metric and the empirical confusion rate. SCS outperforms all baselines on FewRel; the margin is smaller on Few-NERD, where hierarchical naming conventions provide partial signal to string-based metrics. †42% of pairs lack WordNet coverage.

Q Split-Origin Covariate Analysis

FewRel combines 64 train-split and 16 val-split relation types into a single 80-type pool. Train-origin types show numerically lower accuracy (-8.8 pp, $p = 0.06$; see Limitations), raising the question of whether differential pre-training exposure confounds the BCS–accuracy relationship. We test this by adding a per-batch *train-origin ratio* (fraction of relation types drawn from the train split) as a covariate in the BCS–accuracy regression.

Method. Across 213 multi-type batches ($k \geq 2$) from the batch-composition experiment (§5) and the factorial design (§4.3), we compute $\texttt{train_origin_ratio} = n_{\text{train}}/k$ for each batch. The ratio ranges from 0.40 to 1.00 (mean 0.80, consistent with the 64/80 base rate). We fit:

$$\text{acc} = \beta_0 + \beta_{\text{BCS}} \cdot \text{BCS} + \beta_{\text{orig}} \cdot \text{train_origin_ratio} + \varepsilon$$

Results. The BCS partial effect is unchanged after controlling for split-origin: $\beta_{\text{BCS}} = -0.416$, $p < 0.001$ (vs. $\beta = -0.412$, $p < 0.001$ without the covariate). Train-origin ratio is not independently significant ($\beta_{\text{origin}} = -0.057$, $p = 0.245$), and $\Delta R^2 = 0.005$. The BCS–accuracy relationship is therefore not confounded by differential pre-training exposure at the batch level.

Pairwise SCS distributions do not differ significantly between train-origin (64 types, mean = 0.880) and val-origin (16 types, mean = 0.874) subsets (Mann-Whitney $p = 0.754$, Cohen's $d = 0.081$); the train/val split does not introduce systematic confusability bias.

R Cross-Seed Variance Decomposition

Count	Batch SD	Seed SD	Ratio
4	0.0889	0.0122	53.5
8	0.0820	0.0098	70.0
16	0.0626	0.0270	5.4
32	0.0462	0.0091	26.0
40	0.0421	0.0019	474.4
64	0.0245	0.0022	119.8
80	0.0001	0.0088	<0.1

Table 19: Variance decomposition on FewRel (3 seeds $\times$ 25 batches per count level). Batch SD: root-mean intra-seed variance averaged over seeds. Seed SD: standard deviation of per-seed means. Ratio = batch variance / seed variance. At count 80, only one batch composition exists, so batch variance vanishes by construction.

Intra-batch variance dominates inter-seed variance by 5–474$\times$ across count levels 4–64 (mean 125$\times$); random batch composition is the primary source of performance variability. Under greedy decoding (temperature $= 0$), model inference is nearly deterministic; the inter-seed variance reflects differences in which random batch compositions each seed generates, not inference stochasticity. At

count 80, the full schema admits only one composition, leaving residual model-level noise as the sole variance source (seed SD = 0.88 pp, likely attributable to FP8 quantization non-determinism).

S Cross-Domain Connections

The schema-associated degradation documented here is not unique to information extraction. Schema-Guided Dialogue systems (SGD/SGD-X; Rastogi et al. 2020) face an analogous challenge: as the number of intent and slot types grows, dialogue state tracking accuracy declines. Li et al. (2023a) observed count-dependent degradation in ChatGPT-based IE, attributing it to prompt overload. Our two-dimensional decomposition offers a more refined explanation: schema size per se is not the sole predictor; rather, larger schemas increase the probability of including high-SCS pairs, which contribute disproportionately to confusion. This distinction has practical implications: a schema of 80 types with low mutual confusability may outperform a 40-type schema with clustered semantics, and suggests that confusability-aware design principles may generalize to any task where an LLM must discriminate among a prompt-specified label inventory.

T Cross-Model Validation with LLaMA-3.1-70B-Instruct (FP8)

To test whether the two-dimensional degradation pattern generalizes across models, we replicate the count-curve and BCS-controlled experiments on Meta-LLaMA-3.1-70B-Instruct (FP8 quantization). All batch compositions, type pools, and test samples are identical to those used for Qwen3-32B (§4.1, §4.3).

Table 20 shows that RE degradation slopes are near-identical: −30.7 pp (70B) versus −30.1 pp (32B) from count 8 to count 80, with the 70B model approximately 4 pp lower in absolute accuracy at each count level. This quantitative match provides strong evidence that Dimension 1 (count-related decline) is directionally consistent across architectures for RE.

On NER, the count-related decline direction is consistent across models: both 70B and Qwen3 show a recall decline from count 2 to count 22 (70B: 0.323→0.252, −7.1 pp; Qwen3: 0.515→0.279, −23.6 pp). Both models also exhibit a U-shaped rebound at higher counts (70B: 0.252 at count 22 → 0.297 at count 44; Qwen3: 0.279 at count 22 → 0.407 at count 66), which cross-validates the U-shape phenomenon. The magnitude difference reflects a precision-recall tradeoff: 70B operates more conservatively (higher precision, lower recall), while Qwen3 over-predicts (higher recall, lower precision, especially at low counts).

Count	LLaMA-70B	Qwen3-32B
RE (FewRel, Classification Accuracy)		
8	0.757	0.805
16	0.685	0.719
32	0.601	0.652
64	0.504	0.539
80	0.450	0.504
NER (Few-NERD, Text+Type Match Recall)		
2	0.323	0.515
6	0.300	0.399
12	0.278	0.318
22	0.252	0.279
44	0.297	0.309
66	0.273	0.407

Table 20: Cross-model count curves for LLaMA-3.1-70B-Instruct (FP8) and Qwen3-32B. RE decline (count 8→80): −30.7 pp (70B) vs. −30.1 pp (32B). NER decline (count 2→22): −7.1 pp (70B) vs. −23.6 pp (32B); both models exhibit a U-shaped rebound at higher counts. 25 batches per count level, same samples and batch compositions across models (seed 42). All metrics are micro-averaged across batches.

For the confusability dimension (Dimension 2), at count 16 on FewRel, LLaMA-70B achieves low-BCS accuracy of 0.708 versus high-BCS accuracy of 0.599 ($\Delta = +10.9$ pp, 95% CI [+3.0, +20.1]). The confidence interval excludes zero; the confusability effect is thus confirmed at the 70B scale.

In summary, FewRel RE provides a quantitative match across models (near-identical slopes for Dimension 1 and a significant Dimension 2 effect). NER provides directional consistency: the count-related decline direction is preserved and the U-shaped rebound replicates, though magnitudes vary with each model's precision-recall operating point. Together, these results indicate that the degradation pattern is directionally consistent across architectures.

U Cross-Model Validation with Gemini 2.5 Flash (API)

We validate both dimensions on a proprietary model, Gemini 2.5 Flash (Gemini Team, 2025), accessed via API. This extends the cross-model evidence from open-weight families (Qwen, LLaMA) to a proprietary architecture with unknown train-

ing details. We replicate the FewRel count-curve and BCS-controlled experiments using the same 960 balanced samples, batch compositions, and evaluation protocol as Qwen3-32B.

Count	Gemini Flash	Qwen3-32B
1	0.827	—
4	0.873	0.870
8	0.820	0.805
16	0.787	0.719
32	0.739	0.652
64†	0.637	0.539

Table 21: Cross-model count curves for Gemini 2.5 Flash (API) and Qwen3-32B on FewRel RE (classification accuracy). 25 batches per count level, same samples and batch compositions (seed 42). Gemini's count-1 result is included for completeness; Qwen3 count-1 was not collected. Both models show monotonic decline from count 4 onward ($\rho = -1.0$ for both on counts 4–32). †Count 64: Gemini based on 11/25 batches (partial).

Table 21 confirms that the count-related accuracy decline replicates on Gemini 2.5 Flash. From count 4 to count 32, both models show perfectly monotonic decline (Spearman $\rho = -1.0$). Including count 1, the Gemini count curve yields $\rho = -0.900$ ($p = 0.037$); including partial count-64 data (11/25 batches), $\rho = -0.943$ ($p = 0.005$). The decline magnitude is smaller for Gemini (-13.3 pp from count 4 to 32) than for Qwen3 (-21.8 pp), consistent with Gemini's higher absolute accuracy providing more headroom before degradation saturates. At each count level, Gemini matches or exceeds Qwen3, with the gap widening at higher counts (0.3 pp at count 4 vs. 8.7 pp at count 32, 9.8 pp at partial count 64); Gemini appears more robust to schema scaling but not immune.

Dimension 2: BCS-controlled confusability. We replicate the BCS-controlled experiment on Gemini 2.5 Flash using extreme-BCS batches at count 8 ($n = 50$; 25 low-BCS, 25 high-BCS) and count 16 ($n = 10$; 5+5).

At count 8, low-BCS batches achieve 90.5% vs. high-BCS 72.9% ($\Delta = -17.6$ pp, 95% CI [15.1, 20.1]). At count 16, the gap narrows to -8.7 pp, but one high-BCS outlier (BCS $= 0.97$, accuracy 59.9%) inflates Pearson r; leave-one-out excluding this point drops r from -0.80 to -0.42 ($p = 0.22$) and ρ from -0.40 to -0.18 ($p = 0.64$), confirming the count-16 correlation is driven by this single leverage point. The count-8 gap is smaller than Qwen3-32B's (-31.2 pp), consistent with ceiling compression: Gemini's higher baseline (90.5% vs. 82.3%) leaves less room for degradation.

Table 22 summarizes BCS-controlled results across all validated models. Both dimensions thus replicate on a proprietary architecture with unknown training details.

Model	Count	Low BCS	High BCS	Δ	95% C
Qwen3-32B	8	82.3%	51.0%	−31.2 pp	
Qwen3-32B	16	82.8%	51.6%	−31.2 pp	
Llama-8B	8	78.1%	51.0%	−27.1 pp	
Llama-70B FP8	16	—	—	−10.9 pp	[3.0, 20.
Gemini Flash	8	90.5%	72.9%	−17.6 pp	[15.1, 20.

Table 22: Cross-model BCS-controlled FewRel RE summary. All models show significant low-vs-high BCS accuracy gaps. Gap magnitude varies with model capacity (ceiling compression), but the direction is consistent across open-weight (8B–70B) and proprietary architectures.

V Few-Shot Prompting and Confusability

To test whether in-context demonstrations mitigate the BCS-associated confusability effect, we evaluate 0-shot, 1-shot, and 5-shot prompting on FewRel 80-relation RE at count 8 using the same extreme-BCS batch design as §4.3. Each condition uses 50 batches (25 low-BCS, mean ≈ 0.36; 25 high-BCS, mean ≈ 0.80), drawn from the pre-constructed batch pool of §4.3. Few-shot examples are sampled from the FewRel training split for each target type.

Condition	Low-BCS	High-BCS	Δ (pp)	95% CI
0-shot	0.682	0.498	+18.5	[+14.3, +22.7]
1-shot	0.754	0.554	+20.0	[+16.3, +23.5]
5-shot	0.715	0.488	+22.8	[+18.9, +26.8]

Table 23: BCS gap under few-shot prompting (FewRel, count 8). Demonstrations improve absolute accuracy on low-BCS batches (+3.3–7.2 pp over 0-shot) but provide inconsistent benefit on high-BCS batches (+5.6 pp at 1-shot, -1.0 pp at 5-shot). The confusability gap widens from +18.5 pp (0-shot) to +22.8 pp (5-shot). All metrics are micro-averaged across batches.

Demonstrations improve absolute accuracy primarily on low-BCS (easy) batches, where type boundaries are already distinct and examples reinforce correct mappings. On high-BCS batches, where types are semantically confusable, demonstrations provide inconsistent benefit (+5.6 pp at 1-shot, -1.0 pp at 5-shot): the model still substitutes similar types at comparable rates. The net

effect is that the confusability gap widens under few-shot prompting (+4.3 pp from 0-shot to 5-shot, bootstrap $p = 0.014$), with all three confidence intervals excluding zero. This indicates that BCS-associated type confusion is a structural property of the schema composition rather than a deficit addressable through prompt engineering. The non-monotonic low-BCS pattern (0.754 at 1-shot vs. 0.715 at 5-shot) may reflect lost-in-middle effects from increased prompt length (Liu et al., 2024).

The 0-shot gap (+18.5 pp) is smaller than the extreme-batch gap reported in §4.3 (+25.5 pp) because the two experiments use different test splits and batch sampling seeds; direct magnitude comparison across experiments is not appropriate.

W Qualitative Error Analysis

Tables 24 and 25 list the ten most frequently confused type pairs on FewRel 80-relation RE and Few-NERD NER. Confusion counts are aggregated bidirectionally across three random seeds (72,000 predictions per seed for RE; single-seed 66-type extraction for NER).

#	Type A	Type B	SCS	N
1	instance of	taxon rank	0.53	875
2	child	father	0.84	700
3	participant of	participating team	0.75	675
4	mountain range	part of	0.57	575
5	mouth of watercourse	tributary	0.65	549
6	child	mother	0.84	500
7	league	member of	0.66	450
8	country of citizenship	country of origin	0.76	400
9	followed by	follows	0.90	400
10	occupation	voice type	0.57	400

Table 24: Top-10 confused relation pairs on FewRel 80-relation RE. N: bidirectional confusion count aggregated over 3 seeds $\times$ 25 batches. SCS: pairwise Schema Confusability Score (Qwen3 embedding cosine similarity). High-SCS pairs (e.g., *followed by*/*follows*, SCS = 0.90) and family-relation pairs (*child*/*father*, *child*/*mother*) dominate. Some low-SCS pairs (*instance of*/*taxon rank*, SCS = 0.53) reflect domain-specific co-occurrence rather than semantic similarity.

#	Type A	Type B	SCS	N
1	person-other	person-politician	0.93	121
2	location-GPE	organization-other	0.83	81
3	art-film	location-GPE	0.69	57
4	building-other	location-GPE	0.85	56
5	building-other	person-other	0.90	43
6	location-GPE	product-other	0.69	39
7	org-gov/agency	person-other	0.85	34
8	person-actor	person-other	0.90	34
9	building-other	location-road/transit	0.76	32
10	location-GPE	org-sportsleague	0.81	30

Table 25: Top-10 confused entity type pairs on Few-NERD NER. N: bidirectional confusion count from 66-type full-schema extraction. SCS: pairwise Schema Confusability Score (Qwen3 embedding cosine similarity). Intra-category confusions (e.g., *person-other*/*person-politician*, SCS = 0.93) and cross-category pairs sharing referent overlap (*building-other*/*location-GPE*, SCS = 0.85) are most frequent. The "other" subtypes absorb diverse entities and confuse with more specific siblings.

X Schema Label Renaming Experiment

To test whether reducing pairwise SCS through label renaming improves accuracy, we conducted a controlled intervention on FewRel 80-relation RE (count 8, Qwen3-32B). We identified 14 high-SCS relation pairs (covering 12 unique relation types, as two types each appear in multiple pairs; mean SCS = 0.87) and replaced one label in each pair with a semantically distinct paraphrase chosen to minimize embedding cosine similarity (all-MiniLM-L6-v2). Examples: *participant of* $\rightarrow$ *event involvement* (SCS: 0.97 $\rightarrow$ 0.72), *follows* $\rightarrow$ *prior entry* (0.90 $\rightarrow$ 0.69), *part of* $\rightarrow$ *integral segment of* (0.90 $\rightarrow$ 0.64). The renaming reduced mean pairwise SCS from 0.87 to 0.67 ($\Delta = -0.20$). We ran the same 50 BCS-controlled batches (25 low-BCS, 25 high-BCS; 96 samples each) under both original (control) and renamed (treatment) label sets.

Renaming hurt accuracy uniformly: both groups showed significant declines (paired t-test, both $p < 0.001$). The number of renamed types per batch correlated negatively with accuracy change ($r = -0.625$, $p < 0.001$). A solo sanity check confirmed that 11 of 12 renamed labels were individually recognizable (accuracy $\geq$ original -5 pp). The results indicate that replacing familiar Wikidata labels with unfamiliar paraphrases disrupts the model's type semantics; this label unfamiliarity effect outweighs any benefit from reduced confusability.

Y Relaxed Matching Evaluation (NER)

To test whether exact-match F1 masks the BCS signal in NER by conflating boundary errors with type errors, we evaluate three progressively relaxed matching criteria on the full Few-NERD NER dataset ($n = 450$ batches: 6 count levels $\times$ 25 batches $\times$ 3 seeds).

A per-error-mode analysis confirms this null ex-

Sentence	Gold	Pred.	SCS
… played for **Valencia** … [**EuroLeague**]	participant	participant of	.97
Pilões is a municipality in **Rio Grande do Norte** …	contains a.t.e.	located in a.t.e.	.92
Davies … appointed to **Cabinet** as **Sec. of State**	part of	member of	.90

Table 26: Illustrative confusion instances from FewRel 80-rel RE. All gold-predicted pairs have SCS > 0.90: *participant* vs. *participant of* reverses directionality; *contains* vs. *located in* inverts containment; *part of* vs. *member of* conflates meronymy with membership. a.t.e. = admin. territorial entity.

	Control	Treatment	Δ (pp)
Low-BCS ($n = 25$)	90.1	87.1	-3.0
High-BCS ($n = 25$)	69.3	60.5	-8.8

Table 27: Accuracy (%) under original vs. renamed labels on FewRel count-8 batches. Renaming reduced embedding similarity but decreased accuracy in both groups, with a larger drop on high-BCS batches (Cohen's $d = -2.92$) than low-BCS ($d = -1.09$). Interaction: $\Delta_{\text{high}} - \Delta_{\text{low}} = -5.9$ pp ($t = -7.23$, $p < 0.001$).

Match mode	r	p	ρ	p	F1
Exact	0.014	0.760	-0.101	0.033	0.304
Relaxed span	0.003	0.946	-0.110	0.020	0.346
Token overlap	0.004	0.936	-0.113	0.017	0.349
Type-ignored	0.023	0.621	-0.097	0.039	0.425

Table 28: BCS–F1 correlation under progressively relaxed NER matching. Relaxed span credits a match if the predicted span overlaps the gold span and the type is correct; token overlap uses token-level intersection; type-ignored drops the type constraint entirely. Relaxed matching improves mean F1 by 4.2 pp (98.4% of batches improve), but Pearson r remains near zero across all modes (bootstrap 95% CI crosses zero in all cases). Boundary errors affect all batches uniformly and do not mask a BCS signal.

tends to all individual categories: miss ($\rho = 0.017$), boundary ($\rho = -0.043$), and type substitution ($\rho = -0.010$) rates show no BCS association (all $p > 0.3$, $n = 450$). The null holds within each count level (count 2: $r = 0.022$; count 6: $r = 0.110$; count 12: $r = 0.028$; all $p > 0.3$), ruling out count as a confound.